

Multi-resolution audiovisual representation of sound fields recorded with a 7th-order ambisonic microphone array.

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Figure 1: Acoustic power map of a 3rd-order ambisonic recording layered over a 360° picture.

ABSTRACT

As part of the development of an ambisonic microphone array, it is of interest to determine the spatial accuracy required to capture and reproduce a sound field over headphones. A 7th order ambisonic microphone array consisting of 480 MEMS capsules has been developed to capture sound fields with high resolution. This research demonstration allows us to experience the drawbacks of reducing the spatial accuracy of an ambisonic representation obtained with such a prototype. It allows to listen to the captured sound field at different resolutions and to visualize the resulting accuracy in terms of energy over the sphere. The visual information is overlaid on 360 videos to help identify the sound sources around the listener. Such development can find applications in the context of remote monitoring and will enable the study of interactions between auditory and visual cues in the context of Virtual Reality experiences.

Index Terms: Ambisonics, Microphone array, Spatial sound perception, Sound field visualization

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1 INTRODUCTION

Numerous methods have been developed to capture and reproduce sound fields that are nearly indistinguishable from natural hearing, to mimic real-life experiences. Ambisonic microphone arrays play a key role in these efforts by capturing sound fields with physical accuracy. They are widely used for the analysis of spatial sound and the creation of virtual or augmented reality content. Such arrays are designed to use the ambisonic formalism to represent sound fields in the Spherical Harmonic (SH) domain. This approach enables a dynamic binaural rendering of auditory scenes through headphones with head-tracking capabilities. However, the fidelity of the SH representation depends on the physical properties of the microphone array, such as its size, the number of sensors, their placement and directivity, as well as the presence of a baffle. All these characteristics determine the spatial resolution of the recorded sound field over frequency in the SH domain [5].

The physical accuracy of the sound field reconstruction is determined by the ambisonic order, which is the maximal SH order used for its decomposition in the SH domain. The limitation of ambisonic order can significantly affect the spectral content of a sound scene rendering, reducing the localization accuracy and the externalization [1].

This research demonstration allows us to adjust the spatial resolution of a sound field recorded using a 7th-order ambisonic microphone array. The user can experience the influence of the ambisonic order truncation through both visual and audio renderings of the sound field. The demonstration uses a Head Mounted Display (HMD) to provide dynamic binaural playback, as well as a representation of the acoustic power map layered over the 360° video of an audiovisual scene, as depicted in Fig. 1.

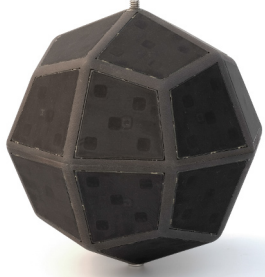


Figure 2: Picture of the 7th-order microphone array used for the sound field recordings.

2 7TH-ORDER AMBISONIC MICROPHONE ARRAY

The ambisonic microphone array was manufactured by Feichter Audio and is composed of 480 MEMS capsules positioned over a 8 cm-large deltoidal icositetrahedron, a polyhedron composed of 24 identical faces. According to the datasheet, the SNR of one capsule is 72 dBA at 94 dB SPL and 1 kHz. The MEMS capsules were assembled by groups of four capsules whose signals were summed to gain about 6 dB of signal-to-noise ratio, *i.e.* leading to 78 dBA of SNR for each channel. Hence, a recording results in 120 signals that are then encoded in 64 audio signals to create a 7th-order representation of the sound field in the ambisonic domain.

The minimum order N that provides a representation of the sound field with a mean squared error of 4% in a sphere of radius r is estimated by $N = \lceil kr \rceil$ [5] where k is the wave number and $\lceil \cdot \rceil$ the ceiling function. Hence, ambisonic components of orders $N > kr$ can be considered to have a negligible influence. Thus, the microphone array and the encoding filters were designed to satisfy this equation with $r = 8.75$ cm, *i.e.* the radius of a large head.

The acoustic scattering induced by the non-spherical baffle of the microphone array was modeled by numerical simulation using NumCalc [3]. The encoding filter computation was then performed using a Tikhonov regularization as described by Moreau *et al.* [4]. The maximum amplification gain of the encoding filters was set to 12 dB.

At high frequency, in order to reduce aliasing-induced spectral coloration, the estimated components were equalized beyond the spatial aliasing frequency in the same way as in [6]. The resulting aliasing frequency was approximately equal to 6.5 kHz.

3 ACOUSTIC POWER MAP

The MVDR method was used to calculate the acoustic power map [2]. This method aims to minimize the contribution of noise or any signal that arrives from directions other than the look direction $\Omega_0 = (\theta_0, \phi_0)$, while maintaining a fixed gain in the look direction. The power spectrum P_{MVDR} in the direction Ω_0 is given by:

$$P_{MVDR}(\Omega_0, k) = \frac{1}{\mathbf{y}_N(\Omega_0)^H \mathbf{R}_{xx}^{-1}(k) \mathbf{y}_N(\Omega_0)}, \quad (1)$$

where $\mathbf{y}(\Omega_0)$ is the steering vector in the look direction Ω_0 composed of the spherical harmonics values up to order N and $\mathbf{R}_{xx}(k) = E[\mathbf{x}(k)\mathbf{x}(k)^H]$ is the covariance matrix of the ambisonic signals $\mathbf{x}$ at frequency k . Hence, the power map is obtained by computing the value of P_{MVDR} in numerous directions around the sphere.

Once computed, the acoustic power maps were superimposed on 360° videos of the scenes. They were recorded in 4K (3840 × 1920

pixels) at 30 frames per second using an Insta360 Pro camera. The videos cover the entire space in both azimuth and elevation.

4 UNITY INTEGRATION

Since the audio clip player of Unity is limited to eight channels, only first-order ambisonic scenes can be used natively. A specific development was undertaken to handle ambisonic files saved in the Opus format [9]. This ambisonic player was used to decode up to 7th-order ambisonic scenes in binaural while taking into account the head rotations of the user. The decoding filters implemented the binaural rendering approach of Schörkhuber *et al.* [8] which optimizes the diffuse-field spectral cues of the decoding filters at different ambisonic orders.

In the application, the user can interact with buttons to change the audiovisual scene as well as the resolution of the sound field representation. All interactions are performed by hand tracking.

5 FUTURE WORK

Future developments include the ability to process an ambisonic stream in real-time to display the acoustic energy map while ensuring the binaural rendering on the fly. Possible applications include industrial or bioacoustic monitoring from remote locations. An additional feature will be the ability to perform beamforming using controllers to isolate or remove a portion of the sound space in order to further analyze the sound field or to edit the recordings.

Moreover, several studies have revealed interactions between auditory and visual cues in the perception of space, with vision often influencing auditory assessments. Following a previous study on the influence of vision over the perception of sound space [7], it is of interest to implement an adaptative procedure in two visual conditions (with and without visual cues) to determine the ambisonic order above which no increase in the spatial auditory accuracy is perceived. It is hypothesized that the presence of vision information limits the need for a high degree of spatial auditory accuracy.

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